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CS101 current midterm papers 2022 Solved By M zaman

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PAPER No. 1

MCQ

1-Both input are 1, output is 0 which logic gate is

Answer: XOR

2-..... retrieve data from CD.?

Answer: Retrieval using a Laser that detects the irregularities on the reflective surface of the CD as it spins

3- The condition in which two or more processes are blocked from progressing because each is waiting for a resource that is allocated to another.

Answer: Deadlock

4-Pop 3 stand for

Answer: Post Office Protocol Version 3 (POP3)

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5-DNS stand.....

Answer: domain name system (DNS)

6- Real time processing

Answer:

Computer performs the tasks in accordance with the deadlines in its external real-world environment. Example of Cruise Missile, Radar etc

7- Example of Software Failure

Fatal Therac-25 X-ray Radiation fatal trace 25 X-ray

Answer:

Fatal Therac-25 X-ray Radiation fatal trace 25 X-ray

8- CPU and main memory are connected by a collection of wires called a

Answer: bus

9- In contrast to an operating system's user interface, the internal part of an operating system is called the

Answer: kernel

10-Secure version of HTTPS is known as

Answer: HTTP

11-protecting login password information technique

Answer: Encryption, decrypt

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12-repetation of the iterative structure is known as

Answer: Loop

Short questions

1-Algorithm informal review in company make a tea (module 85)

Answer:

Making a Cup of Tea

1. Put water in kettle
2. Boil water
3. Put Tea in water
4. Add Milk
5. Stir
6. Add Sugar
7. Put tea in cup
8. Fetch cup

2- Explain DNS and DNS stands for

Answer: The name servers are used as an Internet-wide directory system known as the **domain name system (DNS).** The process of using DNS to perform a translation is called a **DNS lookup.**

3- CPU contain important parts explain...

Answer:

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A CPU consists of three parts: The **arithmetic/logic unit,** which contains the circuitry that performs operations on data (such as addition and subtraction); the **control unit,** which contains the circuitry for coordinating the machine's activities; and the **register unit,** which contains data storage cells (similar to main memory cells), called **registers,** that are used for temporary storage of information within the CPU.

4-

(i) Multi-tasking

Answer:

- ✓ One user executing several tasks simultaneously.
- ✓ Today Multiprogramming is used in single user and multiple users.

(ii) Batch processing

Answer:

The execution of jobs by collecting them in a single batch, then executing them without further interaction with the user.

In batch processing systems, the jobs residing in mass storage wait for execution in a **job queue**

(iii) User interface

Answer:

In order to perform the actions requested by the computer's users, an operating system must be able to communicate with those users. The portion of an operating system that handles this communication is often called the **user interface.** Older user interfaces, called **shells,** communicated with users through textual messages using a keyboard and monitor screen.

(iv) Access-time

Answer: The sum of seek time and rotation delay

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Paper 2

1- Avoid Cyber security Attacks:

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Avoid Cyber-security attacks Avoid searching the followings:

 Free Music The hackers normally add many viruses to such websites as the information seeker of free music etc.

2-shells

Older user interfaces, called shells, communicated with users through textual messages using a keyboard and monitor screen.

3-binary to decimal converter

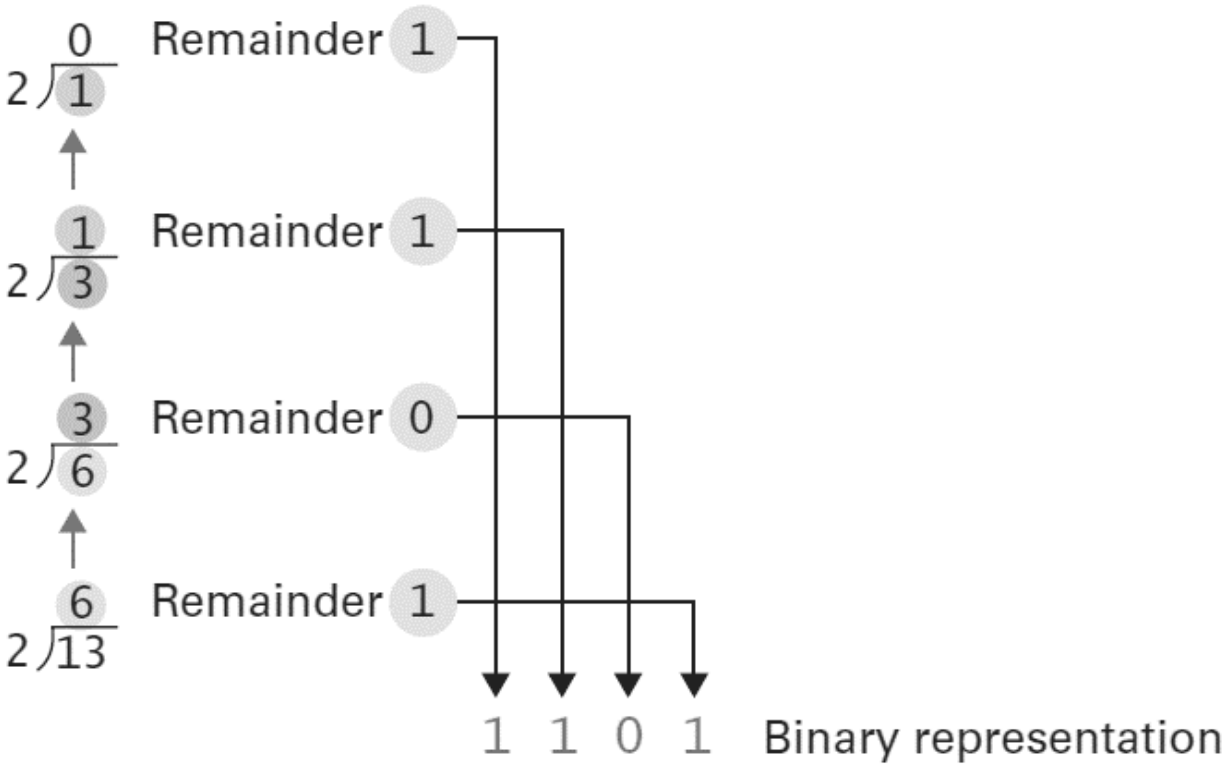
Step 1.	Divide the value by two and record the remainder.
Step 2.	As long as the quotient obtained is not zero, continue to divide the newest quotient by two and record the remainder.
Step 3.	Now that a quotient of zero has been obtained, the binary representation of the original value consists of the remainders listed from right to left in the order they were recorded.

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4-hardware and software embedded application

????????????

Paper 3

Total question 23

Long question with marks 2

Short question 3

Mcqs 18

LONG Question: What is VoIP? Abbreviation of VoIP? And describe using VoIP how we can call someone?

What is VOIP

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✓ Voice over Internet Protocol (VOIP).
✓ Internet infrastructure is used to provide voice communication similar to that of traditional telephone systems

VOIP consists of:
✓ Two processes on different systems using p2p model.
✓ No significant issues in p2p as you might remember

VOIP Challenges:
✓ How to initiate and receive call
✓ Linking VOIP with traditional telephones.
✓ Services like 1122, 15.

✓ Companies consider it as threat to their business, and have made sure to tax them heavily from government or outlawed it completely.

Existing VOIP Systems:
✓ VOIP Soft Phones.
✓ Analog Telephone Adapters
✓ Embedded VOIP phones
✓ Wireless VOIP Technology.

VOIP Soft Phones:
✓ P2P software on both computers without any special hardware.
✓ Skype
✓ Also provide links with traditional telephone communication system.
✓ Drawback: its proprietary, operational standards are not publically known.

2) sidra want to go to mall for shopping? She search on google "the closest shopping mall," and packages of malls Lahore?? Describe sidra searching is right or wrong?

Short:

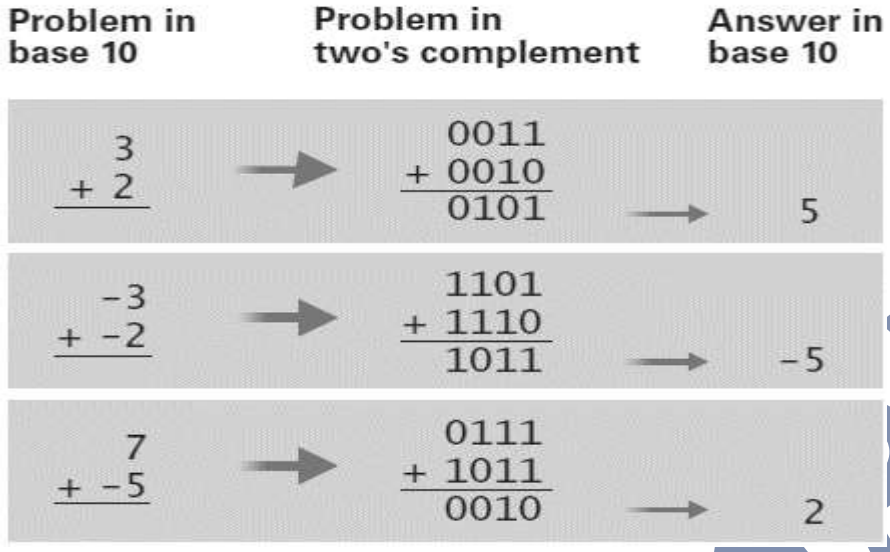
3-Ali want to make a software give him advice to. Make a software?

Write in your own words just describe the benefits of becoming of software developer. It is most demanding job in USA etc.

4-Using 2's floating notation =7 + (-5)

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Paper 4

1-what's the pseudo code of factorial? Long

Let's write a pseudo-code to find a factorial of a number n.

def FindFactorial():

f = 1 i=1; While (i<=n) f=f*i; i=i+1;

Initially "f" and "i" both will have a value of one. The loop will keep executing until "i" remains smaller or equal to n. In each iteration of "i", the current value of "i" will be multiplied with the current value of f.

Let's execute it for n=5:

f=1 i=1

First iteration f= f * I f=1 i=2

Second iteration f= f * I f=2 i=3

Third iteration f= f * I f=6 i=4

Forth Iteration f= f * I f=24

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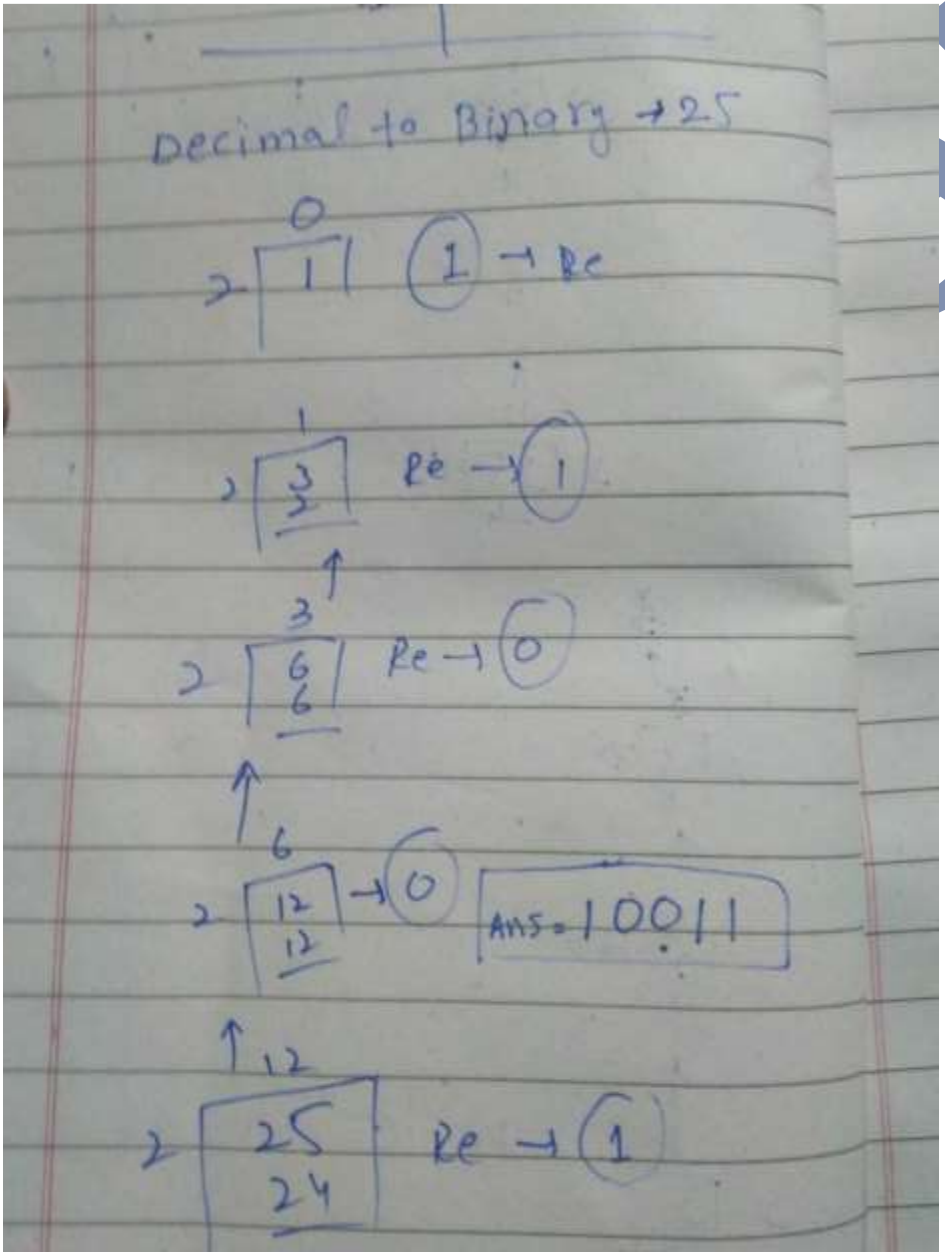
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i=5

Fifth Iteration f= f * I f=120 i=6 when “i” is 6, the loop will terminate as the condition is now false (i<=n)

2-decimal to binary 25



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3-what's internet streaming?

Internet Multimedia Streaming An enormous portion of current Internet traffic is used for transporting audio and video across the Internet in real-time, known as streaming. Netflix streamed more than 4 billion hours of programming to end users in the first three months of 2013 alone. Combined with YouTube, these two services will consume more than half of the bandwidth of the Internet in 2014.

4-The iterative structures

I cannot understand this question so I think they ask to write the names of iterative structures. If you feel something else tell me also.

Answer is

- Sequential Search Algorithm:
- Loop Control
- Components of repetitive control:

5-How many levels of hierarchy?

It have three levels which are given below:

If in short just write names

- Tier-1 ISPs
- Tier-2 ISPs
- Tier-3 ISPs

If long then give details

Individual networks are managed by Internet Service Providers (**ISPs**).

Tier-1 ISP

✓ Relatively few high speed, high capacity, international WANs

✓ Thought of as backbone of the Internet.

✓ Operated by large companies normally originated as telephone company and extended.

Tier-2 ISP

✓ Connecting with Tier-1 ISPs.

✓ More regional in scope, relatively less strong than Tier-1.

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✓ Tier-1 and Tier-2 are network of routers
✓ Forming the core of the Internet.

Tier-3/Access ISP

✓ Independent internet, sometimes referred as intranet, operated by a single authority.
✓ Eg. Cable and telephone companies.
✓ The devices connecting to access ISPs are end systems or hosts.

End systems or Hosts

✓ Laptops/PCs.
✓ Telephones
✓ Video Cameras
✓ Automobiles
✓ Home appliances
✓ The area within the AP or group of APs' range is often called a hot spot.

6-CPU usually uses which pattern for instructions??

The degree to which machine designs should take advantage of this fact has led to two philosophies of CPU architecture. One is that a CPU should be designed to execute a minimal set of machine instructions. This approach leads to what is called a reduced instruction set computer (RISC). The argument in favor of RISC architecture is that such a machine is efficient, fast, and less expensive to manufacture. On the other hand, others argue in favor of CPUs with the ability to execute a large number of complex instructions, even though many of them are technically redundant. The result of this approach is known as a complex instruction set computer (CISC).

MCQS

- **You can rotate the Google screen to 360 degree etc. To do it type the following query:**

  Tilt

- **Excess 4 to represent 100 as Zero**

- **What's a directory**

For the convenience of the machine's users, most file managers allow files to be grouped into a bundle called a directory or folder

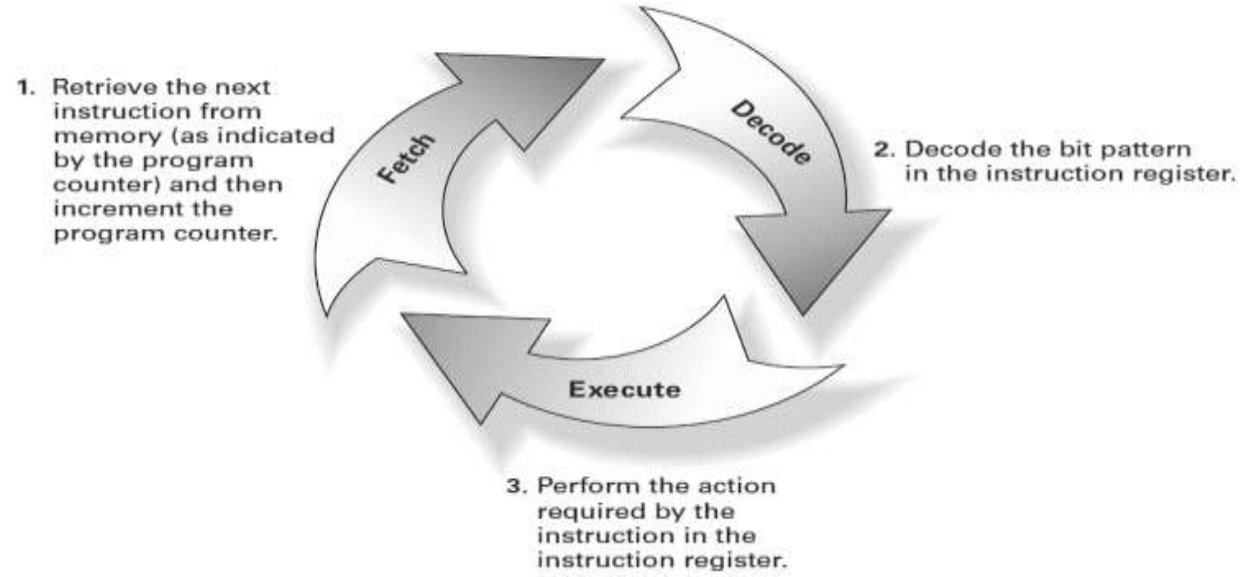
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- **Machine cycle**

3 steps



- **Embedded system**

Still another direction of research in operating systems focuses on devices that are dedicated to specific tasks such as medical devices, vehicle electronics, home appliances, cell phones, or other hand-held computers. The computer systems found in these devices are known as embedded systems.

- **Which point connects to a network**

A bridge is similar to, but more complex than, a repeater. Like a repeater, it connects two buses, but it does not necessarily pass all messages across the connection.

- **Ip address**

Internet needs an internet-wide addressing system that assigns a unique identifying address to each computer in the sys- tem. In the Internet these addresses are known as IP addresses. Originally, each IP address was a pattern of 32 bits, but to provide a larger set of addresses, the process of converting to 128- bit addresses is currently underway.

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Paper 5

1-Which protocol is used to send email??

Simple Mail Transfer Protocol (SMTP)

2-Philosophies name

Two philosophies of CPU

1-RISC

2-CISC

3-Five devices that use are embedded operatory systems?? (5 Marks)

Still another direction of research in operating systems focuses on devices that are dedicated to specific tasks such as medical devices, vehicle electronics, home appliances, cell phones, or other hand-held computers. The computer systems found in these devices are known as embedded systems.

Embedded operating systems are often expected to conserve battery power, meet demanding real-time deadlines, or operate continuously with little or no human oversight

Write uses of these devices in your own words if you want to get good marks.

It's a beautiful tip for you.

4-Write XML code for the given data

Data ni hay to ap is ka concept lain jo bi aye us ka write kar dena

5-Write C78A1 in binary form

Just make a table in which all values are given like.

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Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

C	7	8	A	1
1100	1111	1000	1010	1001

PAPER 6

1-CDN for streaming

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✓ Helps to support the streaming task to neighboring devices.

✓ most of the cases, this is available in Access ISP.

2-Binary to hexadecimal conversion

See module 22

3-ISP levels tree chart

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✓ Video Cameras

✓ Automobiles

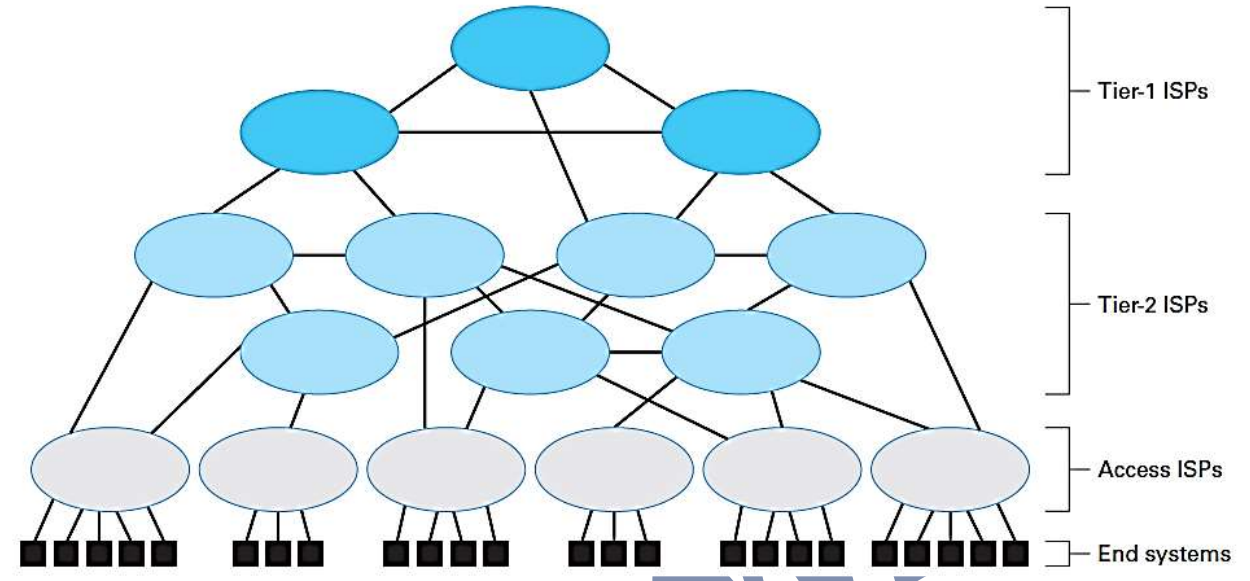
✓ Home appliances

✓ The area within the AP or group of APs' range is often called a hot spot.

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4-HTML program with paragraph tags

Similar to text document, however, just contains some **tags**.

✓ Tags: describe how the content should appear on the screen.

✓ This system of tags is known as HTML (**Hypertext Markup Language**)

Basic structure of HTML document

```
<html>
<head>
<title>demonstration page</title>
</head>
<body>
<h1>Hello</h1>
<p>Introduction to Computing.</p>
</body>
</html>
```

Summary

PAPER 7

1-Artificial intelligence:

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✓ Artificial intelligence is the field of computer science that seeks to build autonomous machines - machines that can carry out complex tasks without human intervention

Applications

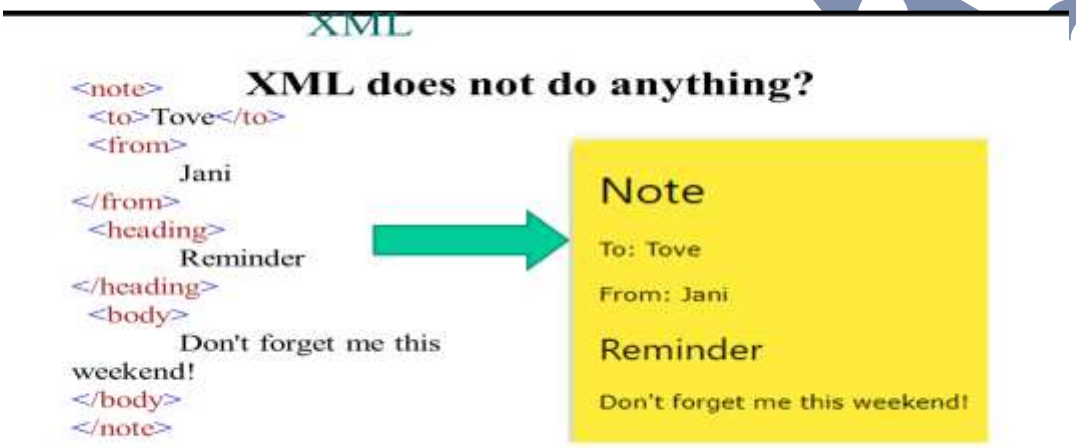
✓ Robots

✓ Self Driving Cars

2-Identify peripheral devices from given devices

Peripheral devices like printer, scanner, and keyboard.

3-XML code



4-Operation Optical System Devices (5 marks)

✓ Compact Disk (CDs)

✓ Digital Versatile Disks **(**DVDs)

✓ Blu-ray Disks (BDs)

Compact Disk

✓ 12 centimeter in diameter

✓ Consists of reflective material

Data Storage on CD

✓ Information is recorded on them by creating variations in their reflective surfaces.

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✓ Data is stored on a single track spiral from inside-outside.

✓ Each track is divided into sectors,

✓ capacity of a sector is 2KB of data, 1/75 of a second audio music.

CD

Data Retrieval on CD

✓ Retrieval using a Laser that detects the irregularities on the reflective surface of the CD as it spins

CD Technology

✓ Initially applied on audio recording using format CD-DA (compact disk-digital audio)

✓ Format is being used even today to store computer data

✓ Single Track

DVDs (Digital Versatile Disks)

CDs capacity is 600 to 700 MB, however, DVD has space in GBs as have multiple semi-transparent layers, which can be viewed by precise focused laser.

BDs (Blue Ray Disks)

BDs use a blue-violet spectrum of light (instead of red), which is able to focus its laser beam with very fine precision. BDs provide five time capacity than DVDs.

5-Five devices that use are embedded operatory systems?? (5 Marks)

Repeated answer is given up

PAPER 7

1-Name popular web browsers

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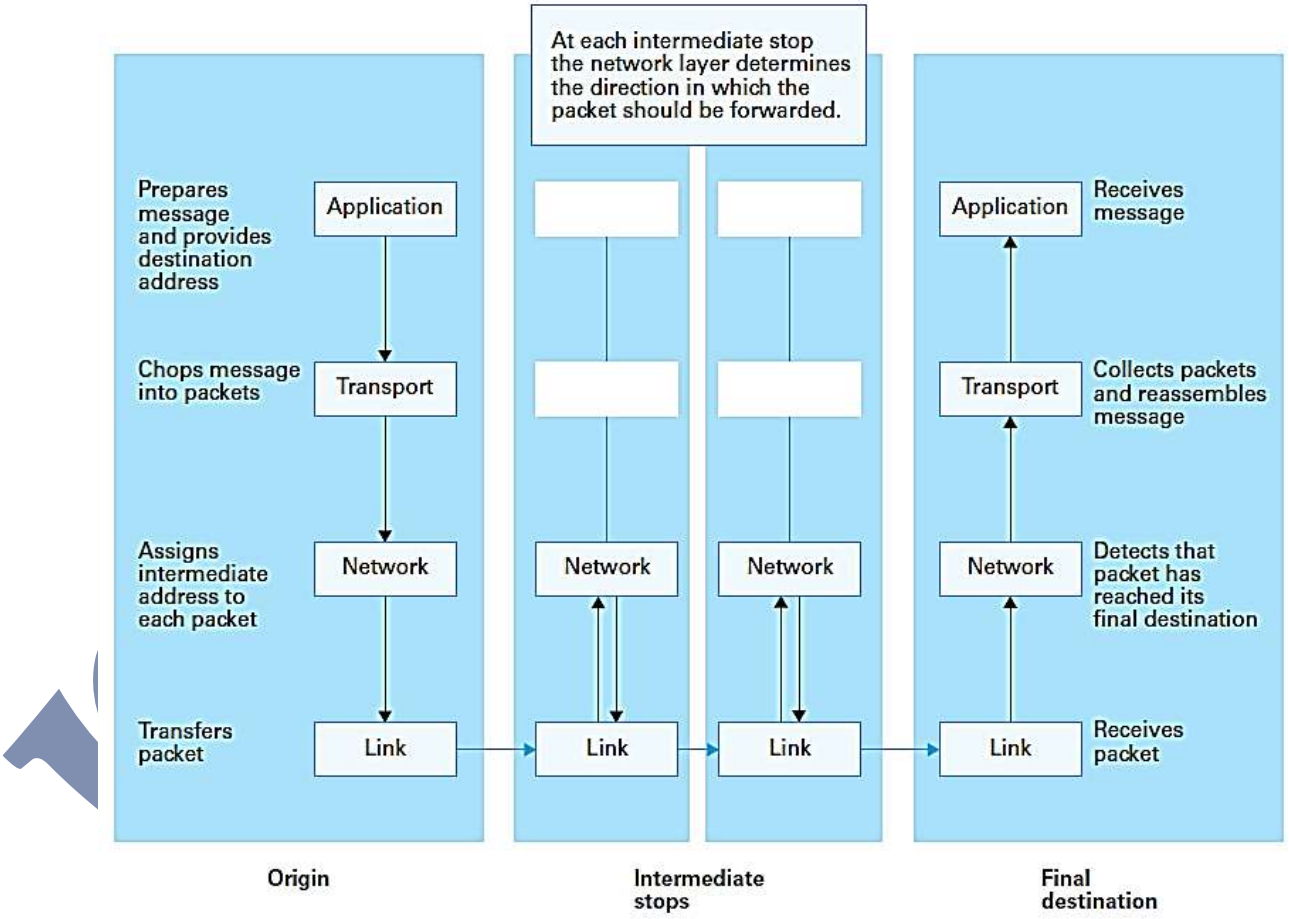
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2- Five devices that use are embedded operatory systems?? (5 Marks) Repeated many times it is important Question.

3- Flow of packets from source to Destination? 5 marks

Module 78



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4- Write a pseudo-code to find a factorial of a number n?? 5 marks

Repeated Question

5- A column was given and just matches the WAN, LAN, PAN, MAN,

Personal Area Network (PAN).

✓ Local Area Network (LAN)

✓ Metropolitan Area Network (MAN)

✓ Wide Area Network (WAN)

Personal Area Network (PAN)

✓ Short range communication (less than a few meters)

✓ Wireless headset and smartphone

✓ Wireless mouse and PC.

Local Area Network (LAN)

✓ Collection of computers in a single building.

✓ Computer in a university campus.

✓ Manufacturing plant.

Metropolitan Area Network (MAN)

✓ Network of intermediate size, one comprising of local community.

✓ Digital network based surveillance system of a city.

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✓ Parking systems interconnected across the city.

Wide Area Network (WAN)

✓ Links machines over a greater distance in neighboring cities.

✓ Connection using internet

6-Which sign is used for Wild Card

“* ”

✓ “* is thicker than water”

7-How many methods are used to analyze a software??

Auditing Software

✓ Administrator can monitor activities.

✓ Malicious or accidental

✓ record and analyze the activities.

Auditing Software - Examples

✓ Flood of attempts to login using incorrect passwords

✓ Impossible Travel.

✓ Violating past behavior

✓ Presence of Sniffing Software.

Sniffing Software

✓ That records user activities and report them to intruder.

✓ An old example is fake simulation of login procedure of OS.

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Quiz:

Kuch quiz k question exactly yad ni hota to ma sara related cheez

apko provide kar rha ho ok

- ✓ Netflix claimed in 2015 that 42.5 billion hours videos have been streamlined by users.

  ✓ In first quarter of 2018, 7.5 million new users have been added.

  ✓ Netflix and Youtube consumed half of Internet Bandwidth.

- ✓ Each Textual Symbol is represented with a unique bit pattern

- Two's complement for storing whole numbers

- ✓ Issues in Unicode for storing numeric values

- Application layer is used to transport of Emails.

Thanks……..

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